

`st.cache` is deprecated and will be removed soon. Please use one of Streamlit's new caching commands, `st.cache_data` or `st.cache_resource`. More information [in our docs](#).

Note: The behavior of `st.cache` was updated in Streamlit 1.36 to the new caching logic used by `st.cache_data` and `st.cache_resource`. This might lead to some problems or unexpected behavior in certain edge cases.

Exploratory Data Analysis of Titanic Dataset

This is an EDA on the Titanic Dataset

First few rows of the dataset:

	PassengerId	Survived	Pclass	Name	Sex	Age	S
0	1	0	3	Braund, Mr. Owen Harris	male	22	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	
2	3	1	3	Heikkinen, Miss. Laina	female	26	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	
4	5	0	3	Allen, Mr. William Henry	male	35	

Missing Values

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0

- ☐ Fill missing Age with median
- ☐ Fill missing Embarked with mode
- ☐ Drop duplicates
- ☐ Drop columns

Cleaned Dataset

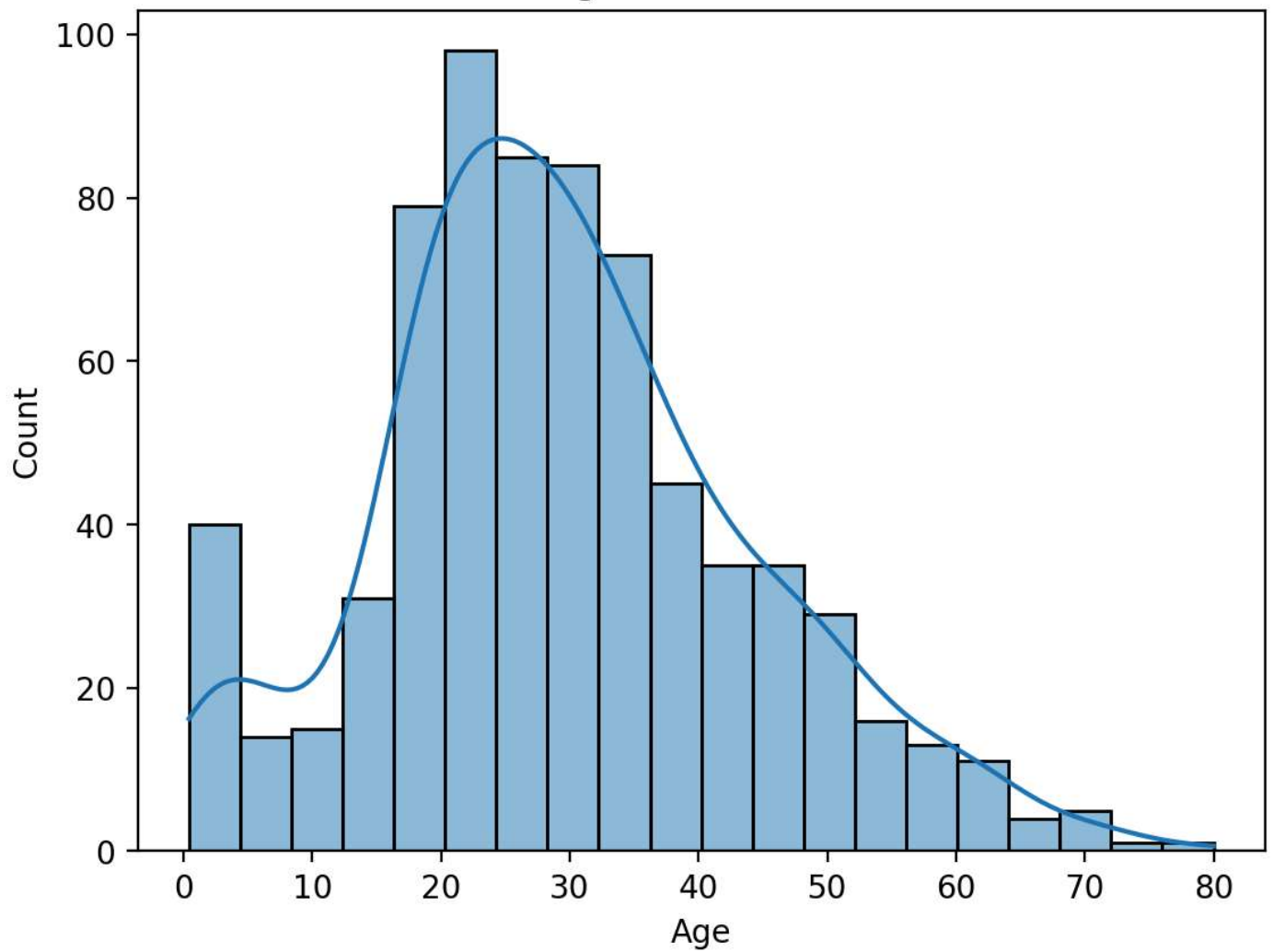
	PassengerId	Survived	Pclass	Name	Sex	Age	S
0	1	0	3	Braund, Mr. Owen Harris	male	22	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	
2	3	1	3	Heikkinen, Miss. Laina	female	26	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	
4	5	0	3	Allen, Mr. William Henry	male	35	
5	6	0	3	Moran, Mr. James	male	None	
6	7	0	1	McCarthy, Mr. Timothy J	male	54	
7	8	0	3	Palsson, Master. Gosta Leonard	male	2	
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	

Statistical Summary of the Data

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891	891	891	714	891	891	891
mean	446	0.3838	2.3086	29.6991	0.523	0.3816	32.2042
std	257.3538	0.4866	0.8361	14.5265	1.1027	0.8061	49.6934
min	1	0	1	0.42	0	0	0
25%	223.5	0	2	20.125	0	0	7.9104
50%	446	0	3	28	0	0	14.4542
75%	668.5	1	3	38	1	0	31
max	891	1	3	80	8	6	512.3292

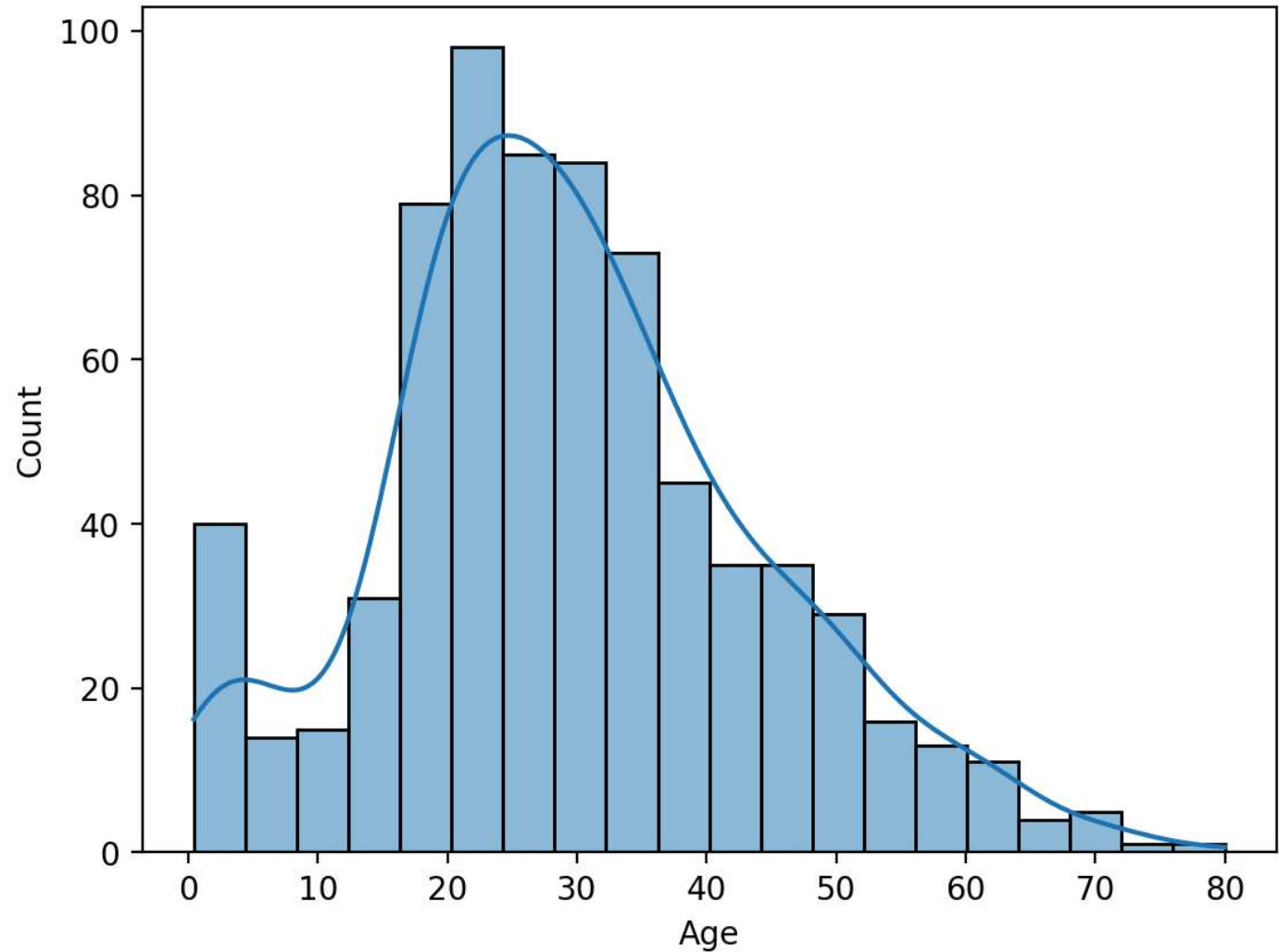
Age Distribution

Age Distribution

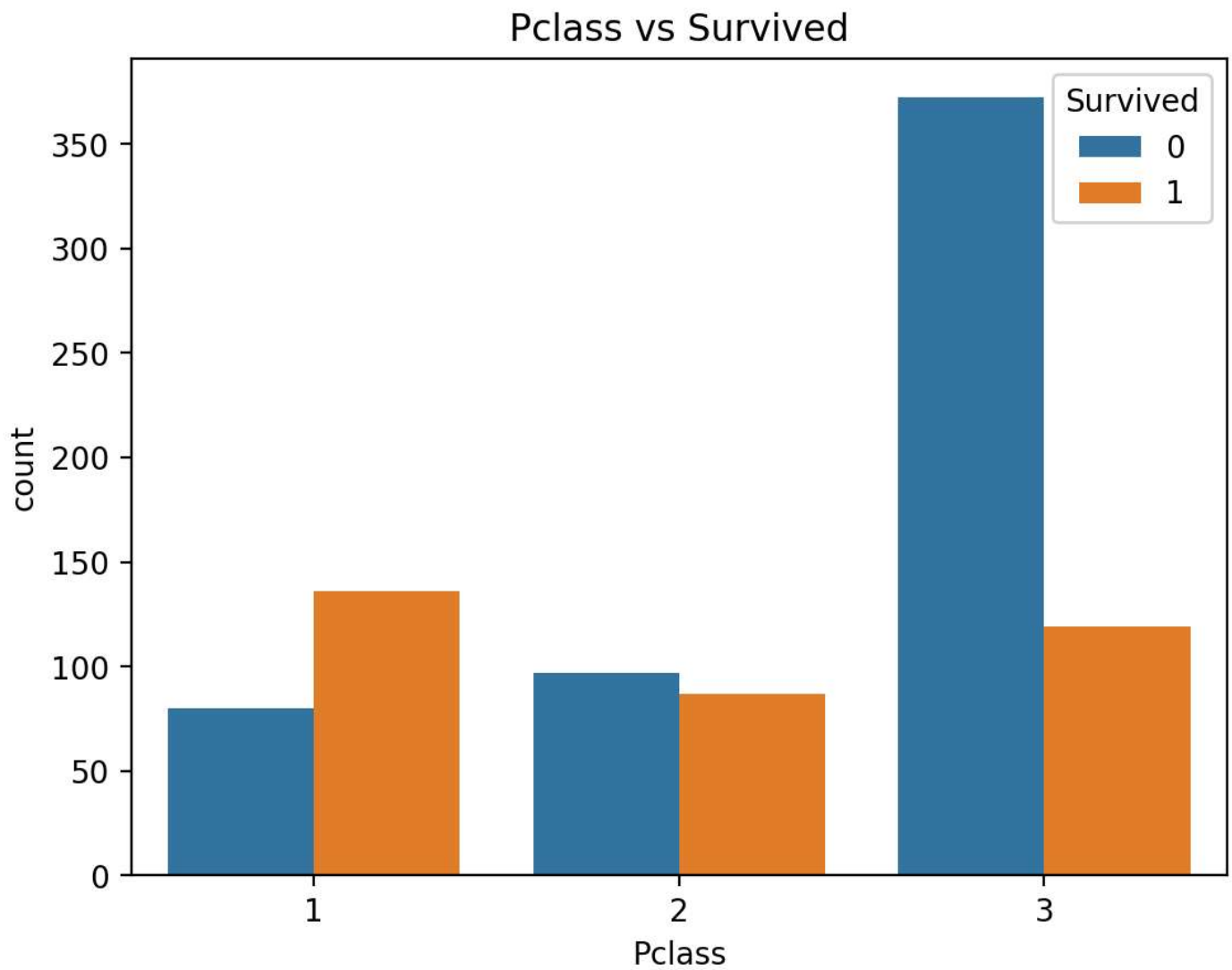


Gender Distribution

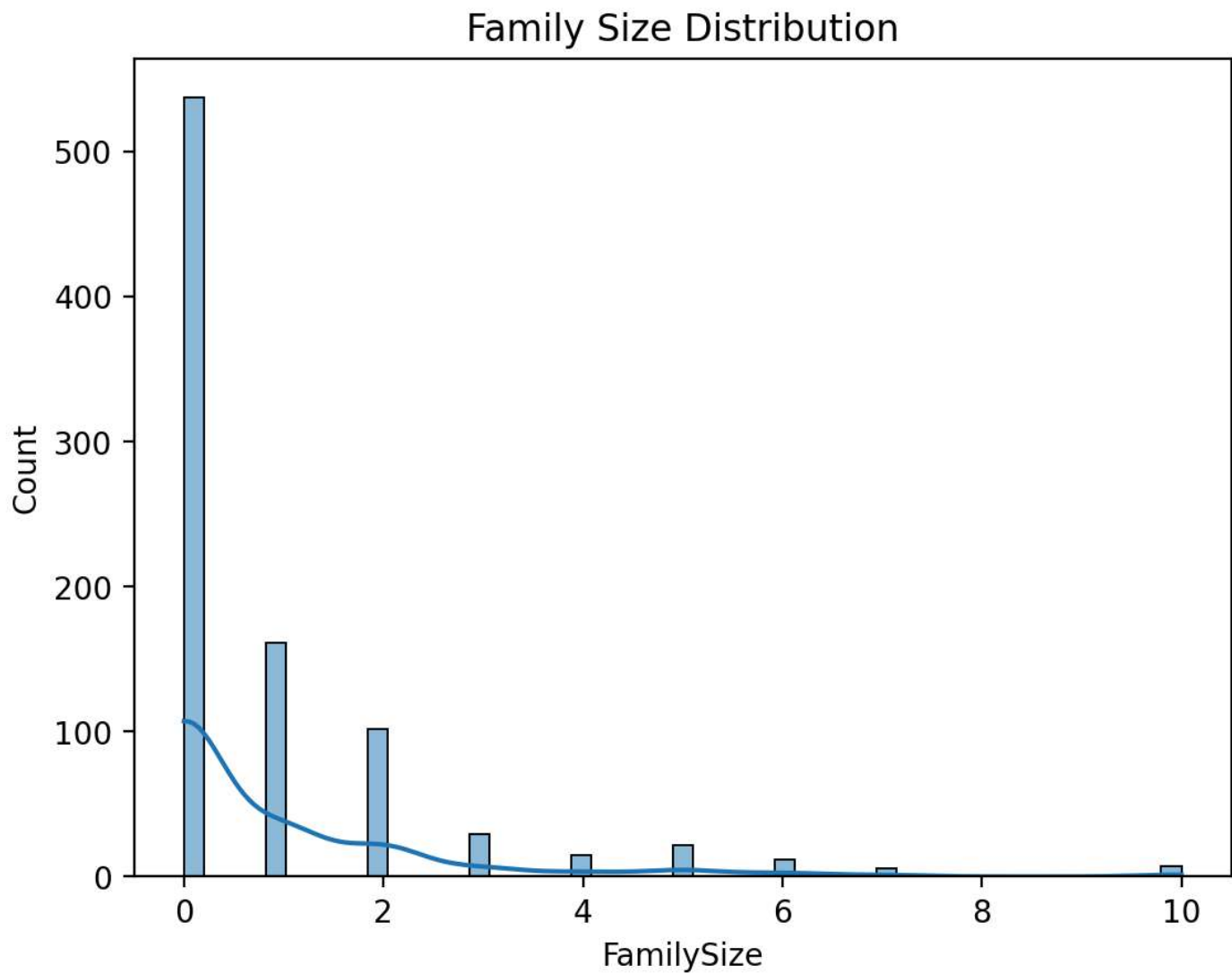
Age Distribution



Pclass vs Survived



Feature Engineering: Family size



Key Insights

- Females have a higher survival rate than males.
- Passengers in 1st class had the highest survival rate.
- The majority of passengers are in Pclass 3.
- Younger passengers tended to survive more often.